

Professional

- **Skill:** Express information effectively to technical and non-technical audiences
Skill Level: Proficient
Evidence: Throughout the course, I developed the ability to explain machine learning concepts—such as supervised learning, overfitting, and neural networks—to both technical peers and those unfamiliar with AI. Reflecting critically, I became more confident in adapting complex information to different audiences.
- **Skill:** Create documents to aid your communication
Skill Level: Trained
Evidence: I wrote detailed reports and summaries of ML experiments, using visuals and statistical charts to support my analysis. These documents enhanced my ability to communicate findings in structured, professional formats.

Commercial Awareness

- **Skill:** Keep current with tools of the industry
Skill Level: Proficient
Evidence: I engaged with cutting-edge tools like Scikit-learn, TensorFlow, and Jupyter Notebook. I also stayed updated on AI trends and ethical issues contributing to my practical and theoretical industry knowledge.
- **Skill:** Seek opportunities to improve and share knowledge of tools
Skill Level: Expert
Evidence: I often helped peers understand ML models and shared useful resources and coding tricks. My willingness to collaborate and exchange tools made me a go-to support figure in practical labs.
- **Skill:** Participate in scientific and professional organisations
Skill Level: Aware
Evidence: I followed relevant ML communities and participated in forums and webinars, helping me stay connected with professional discourse in data science.
- **Skill:** Emphasise quality, customer satisfaction, and fair policies
Skill Level: Trained
Evidence: During projects, I ensured fairness in model predictions by testing for bias and improving data balance, demonstrating ethical awareness in applying ML to real-world problems.
- **Skill:** Demonstrate familiarity with codes of conduct in computing
Skill Level: Proficient

Evidence: I studied ethical ML applications and followed academic guidelines to ensure transparency and compliance, particularly when working with sensitive data.

Subject Understanding, Research, Critical Thinking, Time Management

- **Skill:** Critically analyse complex ideas in Computer Science
Skill Level: Expert
Evidence: I compared and evaluated algorithms like decision trees, SVMs, and deep learning models. I used critical metrics such as precision, recall, and AUC to assess performance, reflecting on their strengths and limitations.
- **Skill:** Recognise inconsistencies and search for additional information
Skill Level: Proficient
Evidence: When models failed or underperformed, I analyzed feature relevance and model parameters, consulted external research, and applied corrections.
- **Skill:** Explore complex real-world problems
Skill Level: Proficient
Evidence: I applied ML methods to real-world datasets (e.g., medical diagnosis and social sentiment analysis), integrating theoretical knowledge with practical applications.

Legal and Ethical

- **Skill:** Comply with laws
Skill Level: Trained
Evidence: I followed data protection standards when handling datasets and ensured that any public data used was anonymised and ethically sourced.
- **Skill:** Maintain privacy and confidentiality
Skill Level: Trained
Evidence: During ML labs, I practiced responsible data handling, especially with user-based datasets, protecting sensitive information.

Social (including Teamwork)

- **Skill:** Act in the best interest of the community
Skill Level: Aware
Evidence: I recognized the societal impact of AI and emphasised fairness and inclusivity when evaluating ML systems and their outcomes.

- **Skill:** Collaborate effectively in diverse teams
Skill Level: Trained
Evidence: Group projects taught me to coordinate tasks, respect differences in opinion, and combine diverse ideas into cohesive ML models.
- **Skill:** Meeting team objectives using teamwork
Skill Level: Trained
Evidence: I contributed effectively to shared goals in lab settings, maintaining team accountability and communication.
- **Skill:** Demonstrate skills in leadership and team building
Skill Level: Trained
Evidence: I often stepped in to clarify coding issues and led discussions on improving model performance, developing informal leadership skills.
- **Skill:** Give and receive constructive feedback
Skill Level: Proficient
Evidence: Peer review was integral to our process, and I improved by incorporating critiques while also guiding others to refine their solutions.

Creativity, Problem Solving, Decision Making

- **Skill:** Create and deliver strategies for sustainability
Skill Level: Aware
Evidence: I examined AI sustainability by looking into energy use in model training and designing efficient architectures where possible.
- **Skill:** Make decisions using multiple information sources
Skill Level: Proficient
Evidence: I triangulated insights from performance metrics, domain literature, and peer input to make model adjustments.

Technical (Data Science)

- **Skill:** Technical skills relevant to degree
Skill Level: Expert
Evidence: I confidently used Python, Scikit-learn, and Pandas for ML implementation. I also explored TensorFlow and Keras for neural networks, applying them to large datasets.
- **Skill:** SQL
Skill Level: Trained

Evidence: Used SQL for data extraction and initial filtering from larger databases before ML analysis.

- **Skill:** Python Programming

Skill Level: Expert

Evidence: I built full ML pipelines using Python, from preprocessing to model tuning and evaluation.

- **Skill:** Java

Skill Level: Aware

Evidence: While the course focused on Python, I reviewed Java implementations of ML concepts to broaden my understanding.

- **Skill:** noSQL

Skill Level: Aware

Evidence: I explored noSQL databases theoretically and noted their relevance in handling unstructured data for ML.

- **Skill:** Scripting Language (Python)

Skill Level: Proficient

Evidence: I wrote reusable scripts for model automation and batch testing.

- **Skill:** Statistical Language (R)

Skill Level: Aware

Evidence: I reviewed R-based statistical methods to compare with Python-based approaches.

- **Skill:** Git repositories

Skill Level: Trained

Evidence: I maintained version control on ML projects using GitHub.

- **Skill:** Use of conferencing technologies and Moodle

Skill Level: Trained

Evidence: I presented findings and communicated with peers via Zoom and Moodle forums.

- **Skill:** Word processing and spreadsheets

Skill Level: Proficient

Evidence: Used Excel for preliminary data analysis and Word for structured reporting.

- **Skill:** Effective use of e-library resources

Skill Level: Proficient

Evidence: Accessed academic databases to research ML theory and real-world applications.

Subject Application

- **Skill:** Take into account other people's perspectives
Skill Level: Proficient
Evidence: Engaged with diverse viewpoints when interpreting model biases and applications.
- **Skill:** Work constructively with differences in viewpoints
Skill Level: Trained
Evidence: Respected diverse technical approaches and opinions during group coding sessions.
- **Skill:** Participate as an informed citizen
Skill Level: Aware
Evidence: Reflected on AI's broader impact on society, advocating responsible use of technology.
- **Skill:** Clarify personal values and ethics
Skill Level: Trained
Evidence: Reflected on fairness, transparency, and accountability in model-building, aligning with ethical standards.